ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

Semester Final Examination
Course No.: Phy 4143
Course Title: Physics II

Winter Semester, A. Y. 2023-2024
Time: 2 Hours
Full Marks: 120

There are 4 **(four)** questions. Answer all 4 **(four)** questions. The symbols have their usual meanings. Programmable calculators are not allowed. Marks of each question and corresponding COs and POs are written in the brackets.

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1. a) Determine the Norton equivalent at terminals *a*-*b* for the circuit in Fig. 1(a). **8 (CO2, PO2)**

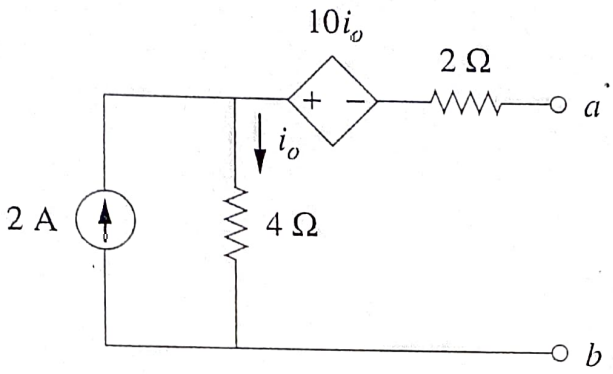


Fig. 1(a)

b) The variable resistor *R* in Fig. 1(b) is adjusted until it absorbs the maximum power from the circuit. **14 (CO2, PO2)**

i) Calculate the value of *R* for maximum power.
ii) Determine the maximum power absorbed by *R*.

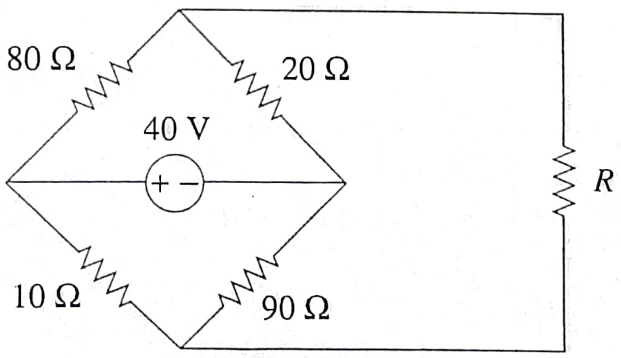


Fig. 1(b)

c) Find i_x in the circuit of Fig. 1(c) using source transformation. **8 (CO2, PO2)**

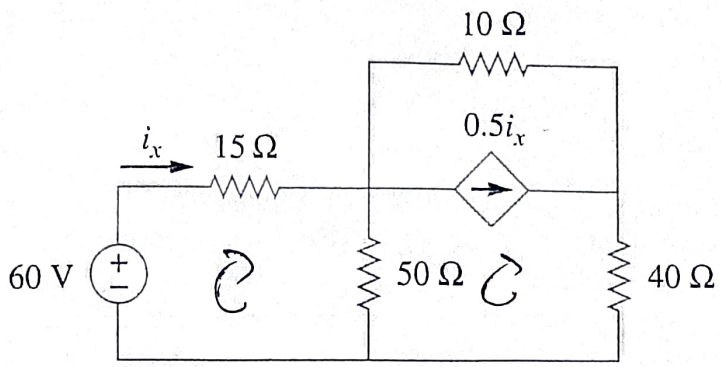


Fig. 1(c)

2. a) Find the Norton equivalent at terminals *a-b* of the circuit in Fig. 2(a). **12 (CO2, PO2)**

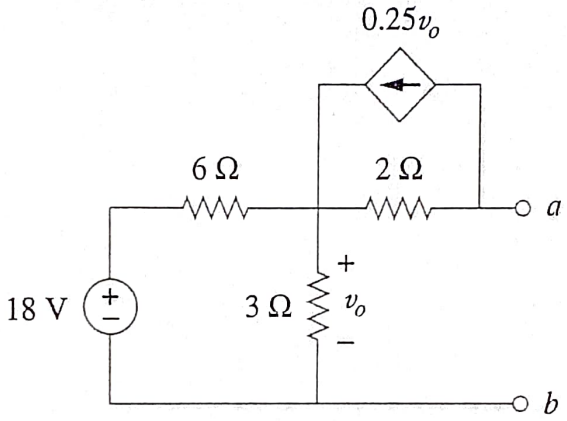


Fig. 2(a)

b) For the circuit in Fig. 2(b), find the Thevenin equivalent as seen from terminals: **18 (CO2, PO2)**

i) *a-b*
ii) *b-c*

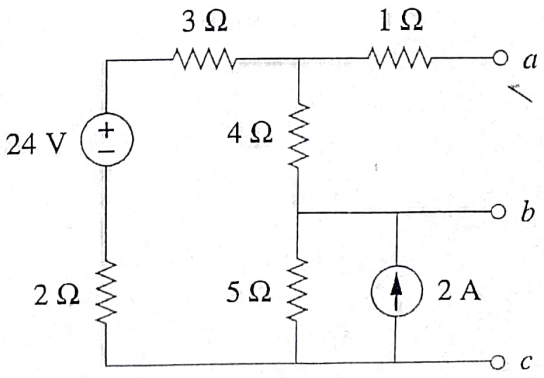


Fig. 2(b)

3. a) The voltage waveform in Fig. 3(a) is applied across a 55 μF capacitor. Describe the voltage waveform mathematically. Sketch the current waveform through the capacitor. **14 (CO3, PO1)**

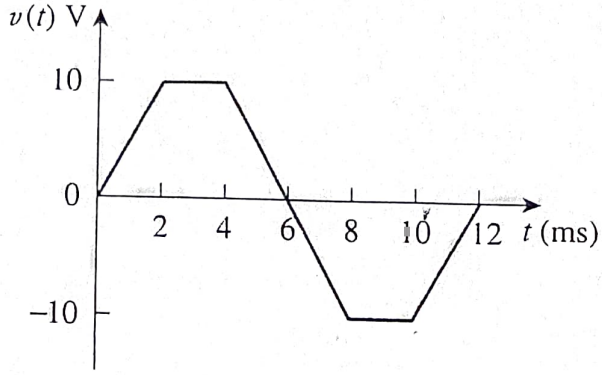


Fig. 3(a)

b) The current through a 5 mH inductor is shown in Fig. 3(b). Determine the voltage across the inductor at $t = 1$, 3, and 5 ms. **8 (CO3, PO1)**

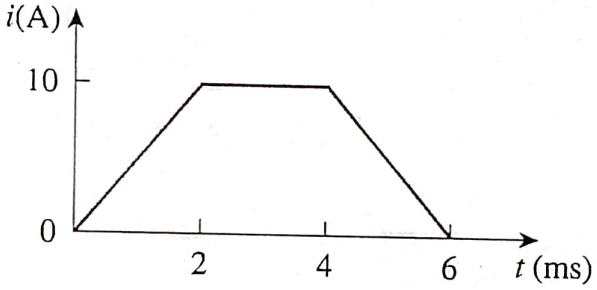


Fig. 3(b)

c) Find v_C, i_L, and the energy stored in the capacitor and inductor in the circuit of Fig. 3(c) under dc conditions. **8 (CO3, PO1)**

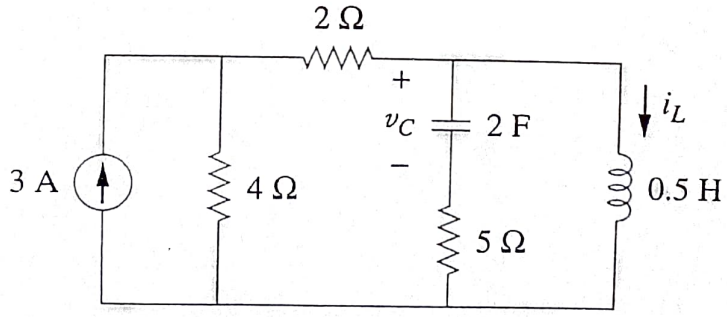


Fig. 3(c)

4. a) Using proper diagrams, justify the statement, "The voltage across a capacitor and the current through an inductor cannot change abruptly" **8 (CO4, PO1)**

b) Calculate $v_o(t)$ in the circuit of Fig. 4(b). **8 (CO4, PO2)**

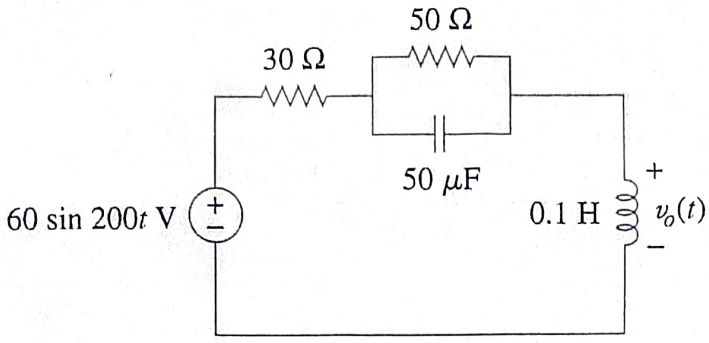


Fig. 4(b)

c) Find current **I** in the circuit of Fig. 4(c). **14 (CO4, PO2)**

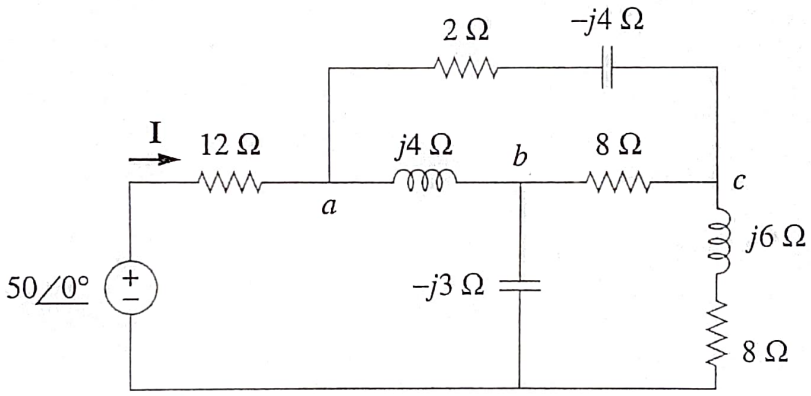


Fig. 4(c)